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AL al-BAYT UNIVERSITY

Faculty Of Prince Al-Hussein Bin Abdallah II for Information Technology

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Dedication

This work is proudly dedicated to:

Our beloved parents, the anchors of our lives, whose unconditional love, endless sacrifices, and continuous prayers have been the cornerstone of our strength and the light guiding us toward this achievement.

Our esteemed supervisor, Dr. Mohammad Alsharo whose profound wisdom, inspiring mentorship, and unwavering support illuminated our academic path and pushed us to strive for excellence.

We dedicate the fruits of this labor to you.

ACKNOWLEDGMENTS

الحمد لله حمدًا كثيرًا طيبًا مباركًا فيه.

Our deepest appreciation and heartfelt thanks go to our parents. We are forever grateful for your endless support, patience, and unwavering belief in us. This milestone would not have been possible without your sacrifices.

We are deeply grateful to our academic supervisor, Dr. Mohammad Alsharo, for his continuous guidance and support. From the beginning of this project to its final stages, his constructive feedback and the time he dedicated to mentoring us have been essential to our success. We truly appreciate his efforts.

We also extend our sincere thanks to the members of our examination committee, Dr. Anas Jebreen and Dr. Manar Mizher, for their valuable time, insightful feedback, and for evaluating our work.

Finally, we thank our siblings, friends, and colleagues who encouraged us along the way and contributed to the completion of this milestone.

Committee Decision

Examination committee

Signature

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This project was submitted in partial fulfillment of the requirements for the Bachelor's degree in Computer Information Systems at the Faculty of Prince Al-Hussein Bin Abdullah II for Information Technology, Al al-Bayt University, Second Semester 2025/2026.

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Abstract

CareerCompass is an interactive web-based academic guidance platform developed by Al al-Bayt University students to address the critical challenge of university major selection faced by high school students in Jordan. The platform serves as a comprehensive digital advisor, bridging the gap between a student's academic potential and future career opportunities through data-driven decision-making. The system utilizes a dual-track logic to cater to two distinct groups: Pre-Tawjihi students (Grades 9–11), who receive actionable improvement plans, and Tawjihi students (Grade 12), who receive immediate university matching. CareerCompass combines an extensive database of local university majors with an intelligent interest assessment test that evaluates students across multiple dimensions. Furthermore, it includes a powerful comparison tool allowing students to analyze and evaluate majors based on required skills, study duration, and career opportunities. Technically, the platform features a user-friendly interface built as a Single Page Application (SPA) with modern web technologies (Vue.js 3 and Tailwind CSS), backed by a robust server-side infrastructure using Python (FastAPI) and SQLite. While currently relying on advanced programmed logic, the system architecture lays the foundation for future Artificial Intelligence (AI) integration. Ultimately, CareerCompass aims to minimize major-switching rates, enhance academic satisfaction, and align with Jordan's vision for digital transformation in higher education, empowering youth to make confident, informed career choices.

CHAPTER 1: OVERVIEW OF THE PROJECT

1.1 Introduction Education is the cornerstone of personal and national development. The transition from high school to university represents one of the most critical decision points in a student's academic journey, shaping not only their next four years of study but also their future career trajectory, earning potential, and overall life satisfaction. In Jordan, this juncture often causes significant stress and confusion, as this foundational choice is frequently made without adequate guidance or reliable information.

In the current educational landscape, students face several persistent challenges that hinder informed decision-making:

- **Information Scarcity:** Comprehensive, reliable information about different majors is often scattered across multiple sources or unavailable.
- **Lack of Personalization:** Generic academic advising often fails to account for individual strengths, interests, and career aspirations.
- **Family Influence:** Students frequently rely on family opinions or social pressure rather than data-driven insights about their own preferences.
- **Comparison Difficulty:** Evaluating multiple majors side-by-side based on meaningful criteria is fragmented and time-consuming.
- **Poor Decision-Making:** The absence of effective pre-enrollment guidance contributes to high rates of major-switching, which wastes valuable time and resources while negatively impacting student satisfaction.

With universities offering an increasingly diverse array of majors, students need a systematic, intelligent tool to navigate these options. With the advancement of web technologies, digital platforms can now offer the personalized guidance that was previously inaccessible. CareerCompass is designed to address these challenges by creating a centralized, interactive platform. It serves as a digital mentor that combines

comprehensive major information, intelligent assessment capabilities, and personalized guidance, effectively aligning a student's academic reality with their personal and professional goals.

1.2 Problem Statement The project addresses several critical issues facing the Jordanian educational sector, where the absence of data-driven guidance has led to a disconnect between student potential and academic reality. The key problems that CareerCompass aims to solve are:

- **Lack of Accessible and Organized Information:** High school students struggle to find comprehensive, reliable information about university majors available in Jordan. Critical details are often scattered across multiple university websites, outdated sources, and unverified forums, making it difficult for students to compare majors or understand career pathways systematically.
- **Absence of Effective Academic Guidance Systems:** Pre-enrollment academic advising is limited or non-existent in many educational contexts. Consequently, students often rely on family opinions or social pressure rather than data-driven insights about their own abilities. Currently, no accessible mechanism exists to scientifically match students with majors based on their personality traits and cognitive preferences.
- **Confusion Due to Major Proliferation:** The large number of available majors (science, engineering, arts, business, etc.) creates decision paralysis and anxiety, particularly during the high-pressure Tawjihi year. Students lack a structured way to narrow down options or understand the distinctions between similar programs, and career outcomes for different majors are often not clearly communicated.
- **Mismatch Between Student Preferences and Chosen Major:** Many students do not have clarity on their own interests, strengths, and career goals before making a major choice. Without objective self-assessment tools, students cannot identify

their academic and professional preferences accurately, leading to dissatisfaction and poor academic performance.

- **High Rate of Major-Switching:** A significant percentage of university students change their major after enrollment due to early poor decision-making. This leads to academic delays, extended time-to-degree, and a waste of financial and educational resources for both families and institutions.
- **Lack of Early Planning (Grades 9-11):** Younger students often reach the critical Tawjihi stage without a clear goal or knowledge of the prerequisite skills needed for their desired fields. The current system lacks tools to help pre-Tawjihi students identify skill gaps early enough to improve them before applying to universities.

1.3 Proposed Solution CareerCompass proposes a comprehensive, integrated web-based platform that serves as an intelligent digital academic advisor. The solution addresses the identified problems by automating the counseling process through a robust, logic-driven architecture. The core components of the solution include:

1. **Dual-Track Logic System:** The platform offers tailored workflows to serve distinct user groups:
 - **Pre-Tawjihi Track (Grades 9-11):** Focuses on early planning by generating custom Action Plans that identify skill gaps and suggest specific subject improvements to reach future goals.
 - **Tawjihi Track (Grade 12):** Focuses on immediate decision-making by providing direct university matching based on final stream selection.
2. **Comprehensive & Localized Major Database:** Unlike generic lists, this component provides a detailed repository aligned strictly with the updated Jordanian secondary fields (**Health, Engineering & IT, Humanities, and Business**). For each major, the system provides:
 - Program overviews, curriculum structures, and course requirements.

- Required foundational skills and typical study duration.
 - Career pathways, job titles, salary expectations, and local employment rates.
 - Direct links to faculties and admission requirements for public and private institutions.
3. **Interactive Interest Assessment System:** A multi-dimensional personality and interest test consisting of exactly **42 non-repetitive scenario-based questions** designed to evaluate a student across weighted categories:
 - Holland's RIASEC professional interests (Weight: 1.0).
 - Core functional aptitudes and multiple intelligences (Weight: 1.5).
 - Work style preferences and environmental stamina (Weight: 1.2). The system incorporates intelligent tie-breaker logic activated when primary dimension scores are highly competitive, ensuring precise field assignment.
 4. **Intelligent Recommendation Engine:** A logic-based engine that processes user grades, chosen academic fields, and assessment results to generate personalized recommendations. To prevent choice overload, it strictly filters and presents the **Top 3 Recommended Majors**, calculating customized fit percentages, required minimum GPAs, and generating tailored diagnostic reasoning justifying the specific selection.
 5. **Comparison and Analysis Tools:** To aid decision-making, the platform includes tools that allow students to compare selected majors side-by-side. This includes visual comparisons of career outcomes (job market demand, growth outlook), skill requirements, and time-to-degree.
 6. **User Personalization and Journey Tracking:** A secure user management system ensures a personalized experience. Students can create profiles to save their assessment history, track results, manage favorite majors, and maintain comparison lists for future reference.

7. **Modern Technical Architecture:** The solution is built upon a highly responsive Single Page Application (SPA) architecture using **Vue.js 3** and **Tailwind CSS** for an optimized frontend interface. It utilizes a secure **SQLite** relational database model for efficient data organization and a robust **Python (FastAPI)** server-side logic layer to handle complex scoring algorithms and recommendations, laying the groundwork for future AI integration.

1.4 Project Objectives

- **Primary Goal:** To develop an interactive, data-driven digital platform that empowers Jordanian students to make informed, confident decisions about their university majors by scientifically aligning their personal interests, academic grades, and career aspirations.
- **Specific Objectives:**
 - **Personalized & Scientific Guidance:** To develop an intelligent, logic-based assessment system that accurately measures student aptitudes. This includes providing immediate matching for Tawjihi students and generating actionable improvement plans for Grade 9-11 students to bridge skill gaps early.
 - **Centralized Information Hub:** To consolidate dispersed information regarding Jordanian universities, majors, and admission requirements into a single, accessible repository. This eliminates information scarcity and helps students connect academic choices to real-world career pathways.
 - **Decision Quality Improvement:** To reduce major-switching rates and academic dropout levels by facilitating better initial decisions. The system aims to minimize the anxiety associated with university applications by replacing social pressure with data-driven insights.

- **Advanced Comparison Capabilities:** To enable students to evaluate multiple majors side-by-side based on meaningful criteria such as study duration, required skills, and employment outcomes, overcoming the difficulty of manual comparison.
- **Enhanced User Experience:** To create an intuitive, responsive web interface that presents complex academic data in a clear, student-friendly format, ensuring accessibility across diverse learning styles and devices.
- **Educational Impact & Alignment:** To support Jordan's national vision for educational excellence by equipping students, parents, and academic counselors with reliable tools that enhance overall academic satisfaction and success rates.

1.5 Stakeholders (Beneficiaries) The CareerCompass platform is designed to serve specific groups within the school education sector:

- **Primary Beneficiaries:**

- **High School Students (Grades 9–12):**
 - **Pre-Tawjihi Students (Grades 9–11):** This group utilizes the platform for early academic planning. They benefit from identifying skill gaps and receiving custom Action Plans to align their abilities with future aspirations before reaching the final year.
 - **Tawjihi Students (Grade 12):** As students preparing for immediate university enrollment, they gain clarity on suitable majors matched specifically to their final stream and grades. The platform empowers them to make confident, data-driven decisions, reducing the risk of poor choices.

- **Secondary Beneficiaries:**

- **Parents:** Parents gain access to reliable, centralized information regarding market demands and career outcomes, enabling them to provide informed support and understand the rationale behind their child's academic preferences.
- **Academic Counselors and Teachers:** The platform serves as a supporting utility that reduces administrative workload. Counselors can use the automated assessment reports to provide more qualitative, personalized guidance during advising sessions.
- **Universities & The Higher Education Sector:** Universities benefit indirectly by receiving applicants who are genuinely suited for their chosen programs. On a national level, this supports educational excellence by potentially reducing major-switching rates and improving student retention.

CHAPTER 2: SYSTEM ANALYSIS

2.1 Introduction After identifying the core problem that CareerCompass aims to address and defining the proposed solution, the next critical phase is comprehensive system analysis. This phase involves gathering detailed requirements from stakeholders, understanding data sources and collection methods, and establishing a systematic development methodology.

The system analysis phase provides the foundation for all subsequent development work. Through careful analysis of user needs, functional requirements, data flows, and system constraints, the team establishes clarity on:

- **What the system must accomplish** (Functional Requirements).
- **How the system should behave** (Non-Functional Requirements and System Constraints).
- **Who will use the system and what each user type needs** (User Roles).
- **When each component should be completed** (Timeline and Project Schedule).
- **How the team will organize and manage the development process** (Methodology).

For CareerCompass, we adopted the Agile/Scrum methodology due to its compatibility with evolving user needs, adaptive development processes, and the team's requirement for flexibility as new insights emerge during the development process. This approach aligns with industry standards where projects are impacted by constraints of time and scope, requiring an adaptive and flexible framework to deliver value quickly.

2.2 Data Collection and Requirements Gathering The data collection and requirements gathering process for CareerCompass involved multiple stages, each designed to capture specific information needed to understand system requirements and user needs.

- **Stage 1: Research on Existing Solutions and Market Trends** To establish a baseline for the project, the team reviewed existing academic advising platforms and major comparison tools. This involved analyzing university websites and studying international best practices in academic guidance.
 - **Key Findings:** The research indicated that while international solutions exist, there are limited local platforms providing integrated major selection guidance in Jordan. Most universities still rely on manual advising without digital decision-support tools, confirming a high demand for organized, centralized academic information.
- **Stage 2: Focus Group and User Feedback Research** The team engaged directly with the target audience to validate assumptions. This involved discussions with high school students preparing for university enrollment, as well as academic counselors and teachers familiar with current advising challenges.
 - **Key Feedback:** The feedback highlighted a strong need for data-driven advising tools to reduce workload. Students specifically expressed a desire for centralized major information, interest assessment tools, and the capability to compare majors side-by-side to understand career outcomes better.
- **Stage 3: University and Program Information Collection** The final stage focused on populating the system's database with accurate content. The team gathered comprehensive data on university majors, including specific curriculum details, requirements, and expected outcomes.
 - **Data Collected:** This process consolidated essential information such as program descriptions, curriculum maps, career pathway insights, employment statistics, and specific admission prerequisites for Jordanian universities.

2.2.1 Quantitative Analysis: User Questionnaire Findings

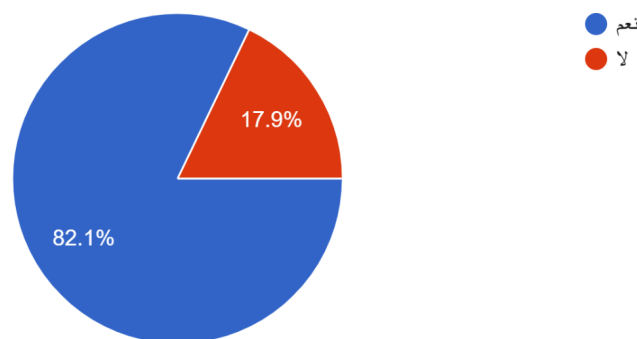
To validate the qualitative findings and deeply understand the magnitude of the problem across different educational stages, we designed and distributed targeted online questionnaires. The surveys captured responses from two distinct groups: high school students currently facing the major selection decision (84 responses) and university students who have already experienced the process (31 responses). The analysis of these 115 responses yielded critical statistics that highlight a systemic gap in academic advising.

The High School Experience: Anxiety and Lack of Tools The data collected from high school students reveals a high-stress environment exacerbated by a lack of professional support:

- **Fear of Misalignment:** A striking 82.1% of surveyed high school students reported feeling afraid of choosing an unsuitable major that fails to meet their ambitions.

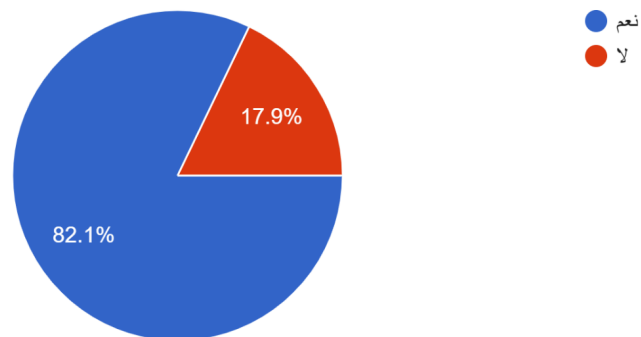
هل تشعر بالخوف من اختيار تخصص غير مناسب أو لا يلبي طموحاتك؟

84 responses



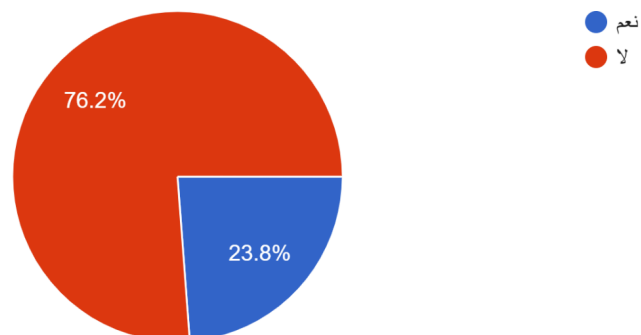
- **Demand for Expert Guidance:** Correspondingly, 82.1% of these students expressed a definitive need for a specialist to guide them in selecting their university major.

هل سبق وأن شعرت أنك بحاجة لشخص متخصص يوجهك في اختيار تخصصك؟
84 responses



- **Absence of Assessment Tools:** Despite this high anxiety and demand for help, 76.2% of the high school respondents had never taken an interest or personality test, either in school or online, to help determine their career path.

هل سبق وأن قمت بإجراء اختبار ميول أو شخصية (سواء في المدرسة أو عبر الإنترنت)؟
84 responses

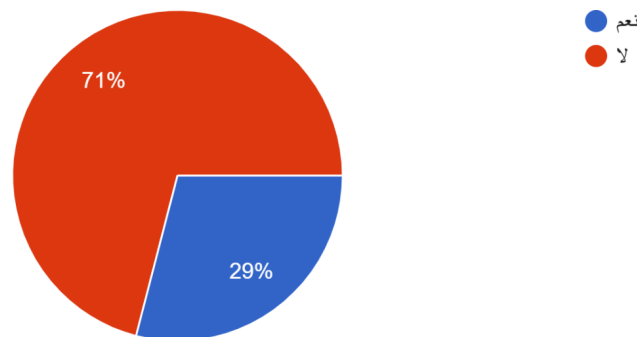


The University Experience: The Cost of Unguided Decisions The responses from university students illustrate the long-term consequences of entering higher education without prior data-driven guidance:

- **Lack of Prior Guidance:** The lack of early intervention is evident, as 71% of university respondents reported that they did not receive any academic or professional guidance prior to selecting their major. Similarly, 77.4% confirmed they had never taken an interest or personality test before making their choice.

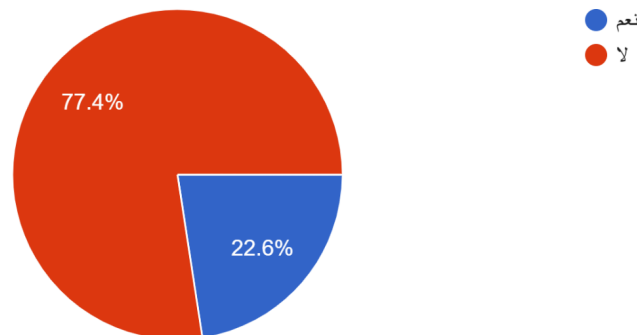
هل تلقيت إرشادًا أكاديميًا أو مهنيًا قبل اختيار تخصصك؟

31 responses



هل سبق وأن قمت بإجراء اختبار ميول أو شخصية (لتحديد مسارك المهني)؟

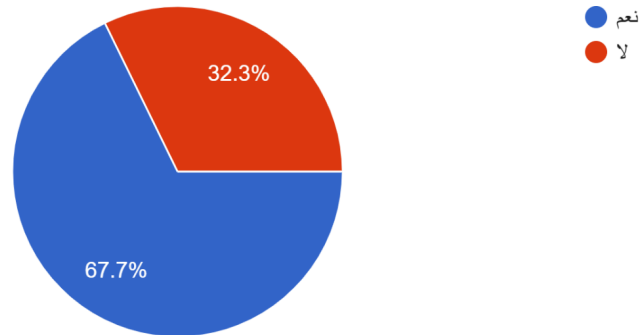
31 responses



- **Future Uncertainty:** The consequences of this unguided selection persist into higher education, with 67.7% of university students indicating they still feel confused or lack clarity regarding their professional future.

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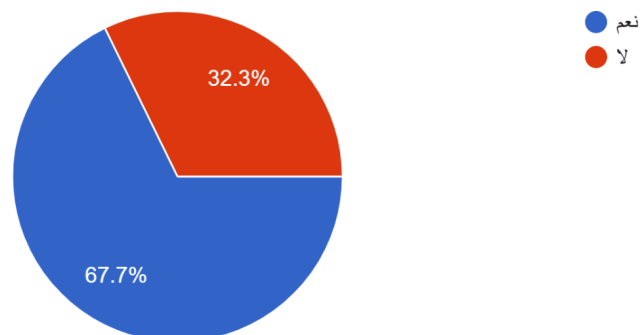
31 responses



- **The Major-Switching Crisis:** Most critically, 67.7% of the surveyed university students admitted to having considered changing their current major after enrollment.

هل فكرت يوماً بتغيير تخصصك؟

31 responses



Conclusion of Findings The statistical evidence gathered from both demographics strongly corroborates our project's problem statement. The data demonstrates a clear timeline of inadequate guidance: high school students experience severe anxiety and lack access to assessment tools, which directly translates into university students suffering from career uncertainty and high rates of major-switching. These findings definitively validate our urgent need to develop CareerCompass as a comprehensive, dual-track digital advising platform designed to intervene early and provide scientifically backed recommendations.

2.3 Methodology We adopted the Agile Scrum framework for its iterative and flexible nature. This approach ensures rapid value delivery and effectively manages evolving requirements throughout the development process.

2.3.1 Adapting Scrum for a Small Team While a typical Scrum Team consists of six to ten members, the framework can be scaled for effective use in different team sizes. As a two-member team, CareerCompass adapted the Scrum Core Roles as follows:

- **Product Owner & Scrum Master (Shared Role):** Both members share the responsibilities of the Product Owner, such as articulating customer requirements and maintaining business justification. Simultaneously, they act as the Scrum Master to facilitate the work, remove impediments, and ensure processes are followed.
- **Scrum Team (Development Team):** Both members function as the cross-functional team responsible for understanding requirements and creating the project deliverables (coding, database management, and testing).

2.3.2 Scrum Principles Applied To ensure project success, the team adhered to key Scrum principles:

- **Collaboration:** Constant interaction was maintained to resolve logic issues in the JavaScript scoring system and ensure shared ownership of the code.
- **Time-boxing:** Work was divided into short, concentrated work cycles called Sprints. Time was treated as a limiting constraint to manage planning and execution effectively.
- **Iterative Development:** The project was built in increments (e.g., Home Page first, then Test Logic, then University Data) to allow for regular feedback and course correction.

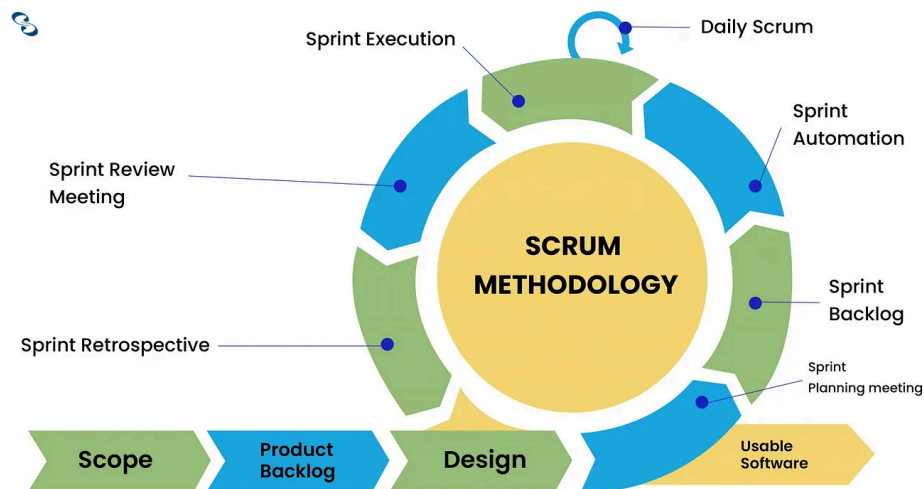


Figure 2.1: The Agile Scrum Methodology Framework

As illustrated in Figure 2.1, the Agile Scrum methodology framework was adopted for the development of the CareerCompass platform. The diagram demonstrates the iterative and incremental software lifecycle, highlighting key phases such as product backlog planning, sprint execution, and continuous feedback loops. This agile approach ensured high adaptability to changing requirements, efficient task management, and the systematic delivery of functional modules at the end of each sprint.

2.3.3 DevOps Integration Practices To enhance the efficiency of our Scrum Sprints and ensure the continuous delivery of a stable web application, the CareerCompass team integrated core DevOps principles into the development lifecycle. DevOps bridges the gap between software development (Dev) and IT operations (Ops), allowing for faster deployment of features and immediate feedback integration.

Our development process followed a continuous iterative cycle tailored to our technical stack:

- **Continuous Integration (CI):**

- **Plan:** User stories and requirements were selected from the Product Backlog at the start of each 2-week Sprint.
- **Code:** Development was conducted using VS Code, utilizing **Vue.js 3**, **Tailwind CSS**, and **JavaScript** for the frontend and **Python (FastAPI)** for the backend logic.
- **Version Control:** Code changes were managed locally to ensure version consistency between team members before merging.

- **Continuous Deployment (CD):**

- **Build & Test:** The application was built and tested locally using **Node.js** and **Uvicorn** server environments. We performed manual Unit Testing on critical components, such as the Assessment Scoring Logic and Recommendation Engine, to ensure calculation accuracy before deployment.
- **Deploy:** Stable versions of the project were deployed to a live hosting environment (e.g., Netlify for frontend prototypes and reliable cloud hosting for the full **Python/SQLite** application). This allowed for real-time performance testing on different devices.

- **Monitor & Feedback:**

- **Operate:** The live system was monitored for availability and responsiveness.
- **Feedback:** Insights gathered from user testing were fed back into the "Plan" phase of the subsequent Sprint, creating a continuous loop of improvement.

2.4 System Requirements This section outlines the specific requirements necessary for the successful development and operation of CareerCompass. These are categorized into Functional Requirements, which define what the system does, and Non-Functional Requirements, which define how the system performs.

2.4.1 Functional Requirements (FR) Functional requirements define the specific behaviors, functions, and interactions the system must support to meet user needs.

- **User Registration & Authentication:** The system must allow students to create personal accounts, log in securely, and manage their profiles.
- **Level Selection:** The system must provide a mechanism for users to select their current academic stage: "Grades 9-11" (Pre-Tawjihi) or "Tawjihi" (Grade 12).
- **Data Input:** The system must accept and validate specific user inputs, including academic grades and major fields (**Health, Engineering & IT, Humanities, and Business**).
- **Assessment Test Logic:** The system must present the interest assessment questions and calculate scores based on predefined logical categories (e.g., **Holland's RIASEC dimensions, Aptitudes, and Work Styles**).
- **Results Generation:** The system must process inputs to display a percentage match for the **Top 3 recommended majors**, along with minimum required GPAs and **tailored diagnostic reasons** justifying each selection.

- **Action Plan Generation:** For Grade 9-11 users, the system must dynamically generate a textual improvement plan identifying skill gaps based on the difference between current grades and future goals.
- **University Search:** The system must provide a searchable and filterable database of localized information regarding Jordanian universities and their programs.

2.4.2 Non-Functional Requirements (NFR) Non-functional requirements specify the quality attributes, performance constraints, and operational standards of the system.

- **Usability:** The user interface must be intuitive, easy to navigate, and culturally appropriate for the target demographic of students aged 14–18.
- **Performance:** The system must optimize logical calculations for test results to ensure they are generated and displayed in under 2 seconds.
- **Responsiveness:** The website design must be responsive, rendering correctly and maintaining functionality across desktop, tablet, and mobile devices.
- **Availability:** The system should ensure high availability (99.9% uptime) via the chosen hosting platform to serve students reliably.
- **Data Integrity:** The database must ensure that user grades, test results, and profile data are stored accurately and remain consistent throughout transactions.
- **Security:**
 - **Data Privacy & Protection:** The system must ensure the secure handling of user data (e.g., emails, academic scores, and assessment results). Since the platform operates on a frictionless model without traditional password-based authentication, privacy is inherently maintained by strictly collecting only necessary academic data without highly sensitive personal identifiers.
 - **Access Control via UUIDs:** Instead of traditional session management and passwords, the system ensures data access security by generating unique, universally unique identifiers (UUIDs) for each submitted assessment. Users

can only retrieve their specific results using these non-sequential, mathematically unguessable IDs, preventing unauthorized browsing of other students' data.

- **Input Validation:** All user inputs (such as grades and text fields) must be strictly sanitized and validated at both the frontend (Vue.js) and backend (FastAPI) levels to protect against common web vulnerabilities, such as SQL Injection and Cross-Site Scripting (XSS).
- **Safety:**
 - **Data Recovery:** The system must include backup mechanisms to prevent data loss in the event of a system failure.
 - **Fail-Safe Operation:** In the event of an error, the system must fail gracefully, providing user-friendly error messages without exposing sensitive system information or crashing the user session.

2.5 Project Timeline

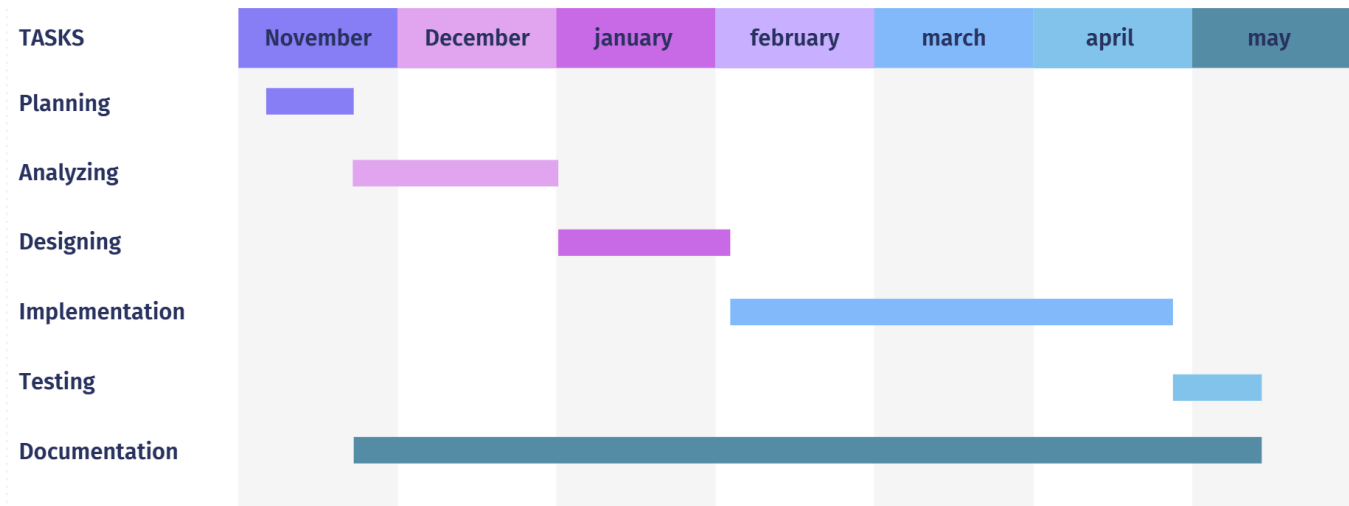
2.5.1 Duration Table

Table 2.1: Duration Table

Tasks	Duration	Start	End
Planning	14 days	2025-11-03	2025-11-20
Analyzing	30 days	2025-11-20	2025-12-31
Designing	24 days	2025-12-31	2026-02-02
Implementation	58 days	2026-02-02	2026-04-22
Testing	17 days	2026-04-22	2026-05-14
Documentation	139 days	2025-11-03	2026-05-14

2.5.2 Gantt Chart

Table 2.2:Gantt Chart



2.5.3 Sprints

The project execution was structured using the Scrum framework, adhering to a consistent **2-week time-boxed iteration** schedule. This consistency allows for regular feedback loops and accurate velocity tracking. Complex phases, particularly the implementation of logic and database integration, were decomposed across multiple Sprints to ensure manageability and continuous delivery of value.

Table 2.3 :Sprints

Sprint	Dates	Phase	Key Deliverables
S1	Nov 3–16	Planning	Project Proposal, Topic Selection & Approval.

S2	Nov 17–30	System Analysis	Initial Data Collection, Market Trends & Existing Solutions Analysis.
S3	Dec 1–14	System Analysis	Focus Groups, Secondary Data (Jordanian Universities & Fields).
S4	Dec 15–28	System Analysis	Functional (FR) & Non-Functional (NFR) Requirements.
S5	Dec 29–Jan 11	Designing	System Diagrams: ERD (Logical Schema), DFD (Levels 0, 1 & 2).
S6	Jan 12–Jan 25	Designing	UI/UX Wireframes, Database Schema Design (SQLite).
S7	Feb 2–Feb 15	System Implementation	Dev Environment Setup, Vue.js 3 Frontend & Pinia State Management.
S8	Feb 16–Mar 1	System Implementation	FastAPI Backend Initialization, SQLite Database Creation, UUID Logic.

S9	Mar 2–Mar 15	System Implementation	Assessment Interface & Heuristic Scoring Logic (Python).
S10	Mar 16–Mar 29	System Implementation	Dual-Track Logic: Recommendation Engine & Gap Analysis (Action Plan).
S11	Mar 30–Apr 12	System Implementation	Frontend-Backend API Integration, Local Server Configuration (Uvicorn).
S12	Apr 13–Apr 26	Testing	Local Unit Testing, Integration Testing, Test Reports.
S13	Apr 27–May 10	Testing	User Acceptance Testing (UAT), Bug Fixing, Performance Optimization.
S14	May 11–May 14	Project Delivery	Final Local Build & System Assembly, Final Documentation Handover.

Chapter 3: System Design

3.1 Introduction Following the comprehensive system analysis and requirement gathering phases, the project transitions into the system design phase. This stage is critical as it bridges the gap between the theoretical requirements and the practical implementation of CareerCompass.

In this chapter, we define the architectural structure of the web application, focusing on how users interact with the system and how data is organized and processed. The design process involves creating visual models that represent the logical flow of information and the relationships between different data entities. By utilizing Data Flow Diagrams (DFD) and Entity Relationship Diagrams (ERD), we establish a clear blueprint that guides the coding and implementation phase. The design ensures that the system is scalable, efficient, and capable of handling the specific needs of both Pre-Tawjihi and Tawjihi students.

3.2 User Roles on the System The system is designed to accommodate different types of users, each with specific access privileges and functional capabilities based on their needs. The platform distinguishes between the following roles:

- **Guest (Unregistered Visitor):**
 - **Access Level:** Limited.
 - **Capabilities:** Guests can access the landing page to view general information about CareerCompass, read the "About Us" section, and use the "Contact" form. They cannot access the assessment tools or receive personalized recommendations without creating an account.
- **Registered Student (Grades 9-11 - Pre-Tawjihi):**
 - **Access Level:** Full User (Planning Track).
 - **Capabilities:**

- **Profile Management:** Create and edit a profile including academic field (**Health, Engineering & IT, Humanities, and Business**) and specific subject grades.
 - **Goal Setting:** Define future career aspirations to tailor the system's analysis.
 - **Assessment:** Take the interest and aptitude test.
 - **Action Plan:** Receive a personalized improvement plan that identifies skill gaps and suggests academic focus areas to prepare for future university majors.
- **Registered Student (Grade 12 - Tawjihi):**
 - **Access Level:** Full User (Decision Track).
 - **Capabilities:**
 - **Profile Management:** Input final or expected Tawjihi grades and academic field.
 - **Assessment:** Take the interest and aptitude test to validate personal inclinations.
 - **University Matching:** View a direct list of the **Top 3 recommended majors** and universities in Jordan that match their specific grade averages and test results, along with **tailored diagnostic reasons**.
 - **Comparison Tools:** Utilize the comparison feature to evaluate different majors side-by-side.
- **Administrator (Admin):**
 - **Access Level:** Backend Control.
 - **Capabilities:**
 - **Database Management:** Add, update, or delete records of Jordanian universities and majors (e.g., updating admission rates or adding new programs).

- **User Oversight:** View registered user statistics and manage user accounts if necessary.

3.3 System Design Diagrams To transform the system requirements into a technical reality, it is essential to visualize the logical structure and data movement within CareerCompass. This section presents the architectural blueprints of the system using standard modeling techniques: Data Flow Diagrams (DFD) and Entity Relationship Diagrams (ERD).

These diagrams serve as the bridge between the analysis phase and the implementation phase. They provide a clear graphical representation of how data is processed, stored, and exchanged between the various components of the system. This ensures that the backend logic, specifically the recommendation engine and user profile management, is structurally sound before coding begins.

3.3.1 Data Flow Diagram (DFD) The Data Flow Diagram (DFD) serves as a visual representation of the information flow within the CareerCompass system. It effectively maps out the journey of data, illustrating how user inputs like academic grades, stream selection, and assessment answers are processed and transformed into meaningful outputs such as major recommendations and personalized action plans. This diagram provides a clear overview of the system interactions with external entities, specifically the students, and details the internal processes and data stores required to execute the core logic of the platform.

3.3.1.1 Context Diagram The Context Diagram represents the highest level view of the CareerCompass system boundaries. It depicts the entire system as a single process interacting with the Student and the Admin as the external entities.

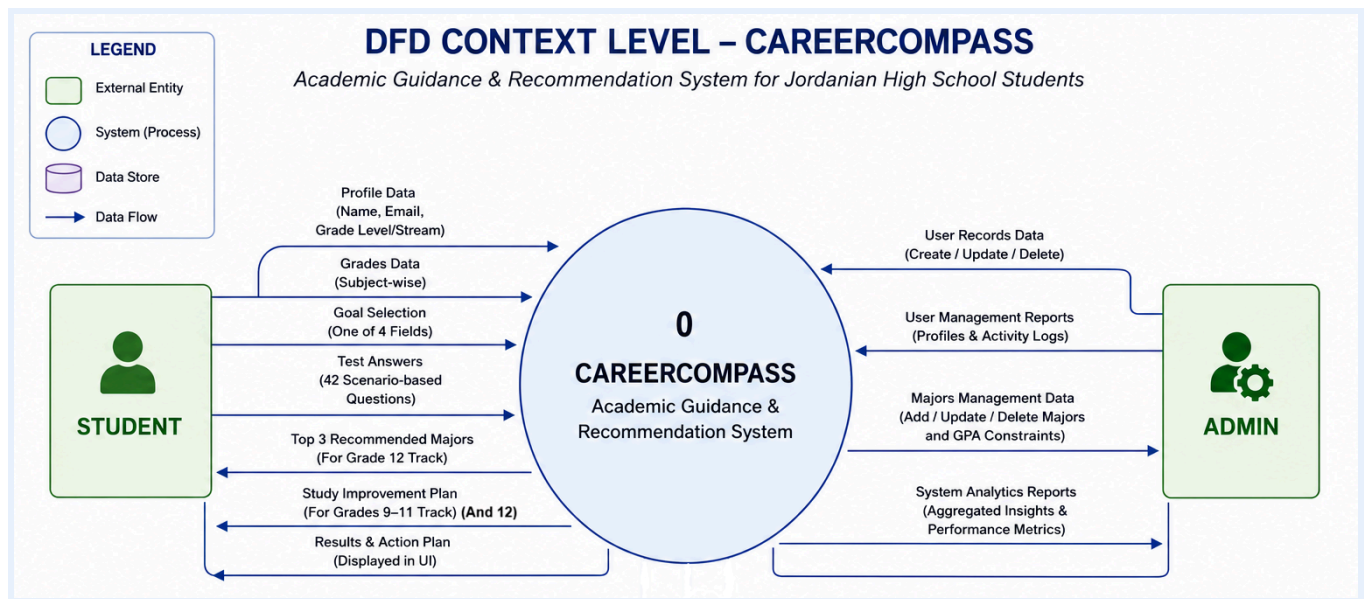


Figure 3.1: DFD Context Level

Figure 3.1 illustrates the Context Level Data Flow Diagram (DFD) for the CareerCompass platform. It presents a high-level overview of the entire system as a single main process, highlighting the primary interactions, inputs, and outputs between the system and its external entities, such as the students and administrators.

3.3.1.2 Level 0 Diagram The Level 0 Diagram breaks down the system into its core functional components: **User Authentication, Academic Data Management, Goal Selection, Interest Test, Recommendation Engine, Study Plan Generator, and Results Presentation**. It highlights the interaction between these primary processes and central data stores, illustrating how student and university data is logically managed and accessed.

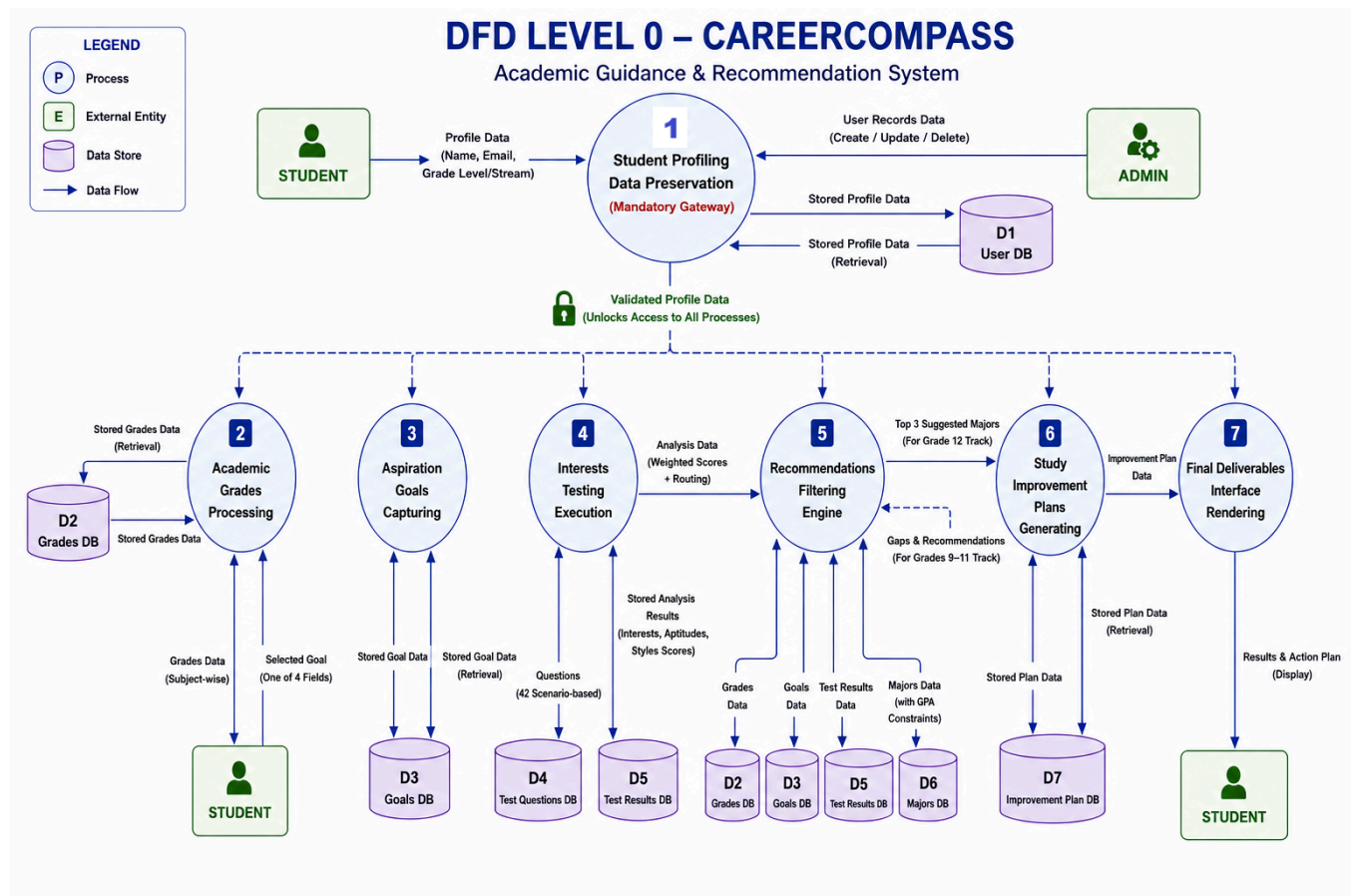


Figure 3.2: DFD Level 0

Figure 3.2 depicts the Level 0 Data Flow Diagram (DFD) for the CareerCompass platform. It illustrates the seven core functional processes, mapping the internal flow of information and detailing how data moves between key modules user authentication, academic grades processing, goals capturing, interests testing, recommendations filtering, study plans generation, and deliverables rendering along with their respective data stores.

3.3.2 Entity Relationship Diagram (ERD) The Entity Relationship Diagram defines the logical structure of the database. It visualizes the entities, such as Users, Grades, Goals, and Universities, along with their specific attributes and the relationships between them. This diagram serves as the blueprint for constructing the **SQLite database**, ensuring data integrity and efficient retrieval of localized Jordanian university information.

CareerCompass ERD (Chen's Notation)

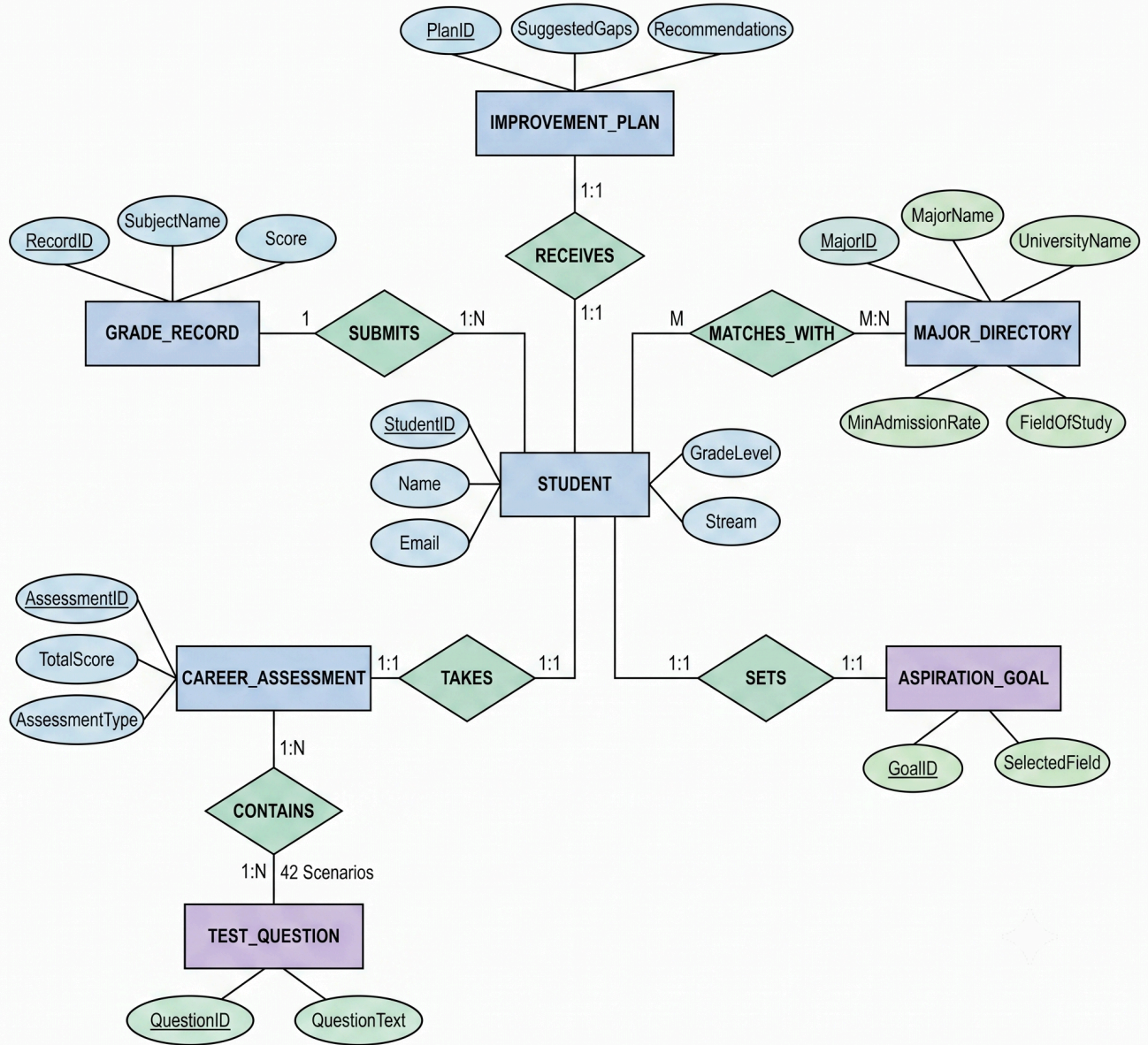


Figure 3.3: CareerCompass ERD (Chen's Notation)

Figure 3.3 illustrates the system's database structure using Chen's ERD notation, mapping the logical relationships between students, their assessments, academic grades, and recommended university majors.

Chapter 4: Software Development and Testing

4.1 Introduction This chapter provides a comprehensive overview of the implementation phase of CareerCompass. It details the modern frameworks used, the development environment, and the rigorous testing procedures followed to ensure the platform's reliability, accuracy in executing weighted multi-dimensional assessment logic, and efficiency in generating precise academic matching plans.

4.2 Development of the Software The development process followed an Agile Scrum framework, focusing on seamless integration between the modular frontend diagnostic interfaces and the robust server-side processing layer. The system architecture enforces a clean separation of concerns, where the client-side handles dynamic user logic, state reactivity, and immediate UI rendering, while the backend layer securely manages persistent storage commits and definitive major database queries.

4.3 Programming Languages and Technologies Based on the finalized implementation, the following technologies were utilized:

- **Frontend Framework:** Engineered as a high-performance Single Page Application (SPA) utilizing Vue.js 3 (Composition API) for highly reactive views. JavaScript (ES6+) implements the local scoring matrices and client logic, combined with Tailwind CSS utility classes to establish a consistent, accessible "Glassmorphism" UI design system.
- **Backend Infrastructure:** Server-side logic and database connectivity are developed using Python within the FastAPI framework, chosen for its high performance and native support for asynchronous API endpoints.
- **Database Management:** SQLite is utilized as the secure relational database management system to store registered student profiles, calculated session results, targeted goals, and the localized Jordanian university repository.

- **Centralized State Management:** Pinia is integrated directly into the Vue ecosystem to securely maintain reactive user sessions, grade profiles, and active assessment logic workflows seamlessly across route transitions.

4.4 Software Tools

- **Integrated Development Environment (IDE):** Visual Studio Code (VS Code).
- **Source Control Management:** Git and GitHub.
- **Runtime Environment:** Node.js (for the frontend execution) and Uvicorn (as the ASGI server running FastAPI).
- **Interface Design and Prototyping:** Figma.

4.5 Testing and Reviewing the Code Testing was executed iteratively across multiple functional levels to guarantee absolute system precision:

- **Unit Testing:** Thoroughly verified the custom Python algorithms responsible for applying multi-scale weights (Interests: 1.0, Work Styles: 1.2, Aptitudes: 1.5) to ensure raw user answers are accurately transformed into valid evaluation dimensions.
- **Integration Testing:** Validated asynchronous requests bridging the Vue.js interface with backend FastAPI endpoints, verifying accurate SQLite database record insertion and error-free querying.
- **Interface and Usability Validation:** Tested responsive Tailwind layouts across desktop screens and mobile viewports, applying meticulous review to ensure flawless Right-to-Left (RTL) formatting and visual readability.
- **Logic Validation:** Manually audited the conditional recommendation engine to confirm that edge-case scenarios trigger the intelligent tie-breaker logic when primary dimension scores show competitive proximity.

4.6 DevOps Principles & Local Build Implementation: In this section, we detail the practical application of DevOps principles that facilitated the continuous development of the CareerCompass platform. While the system operates in a localized environment for this delivery phase, the implementation focused on production-ready technical workflows:

- **Version Control & Collaboration:** We utilized Git for local version control and GitHub as our central repository. This ensured that all code changes from the Vue.js frontend to the FastAPI backend were tracked, preventing conflicts and maintaining a reliable source of truth throughout the development sprints.
- **Continuous Integration (CI):** During the implementation, we maintained a local CI environment using Node.js for the frontend and Uvicorn for the backend. Every time a new feature was added, a local build and manual unit testing were performed to ensure integration stability before the code was finalized.
- **Deployment-Ready Assembly (Local Build):** Instead of a live cloud deployment, we focused on ensuring the system is strictly "deployment-ready." The frontend SPA was compiled into a final production build, and the FastAPI backend was configured to serve requests flawlessly alongside the SQLite database on a local server. This approach allowed for rigorous performance monitoring across different emulated devices without incurring cloud hosting overhead.
- **Monitoring and Feedback Loop:** We implemented a continuous feedback loop where insights from local user interface testing were immediately fed back into the development cycle for the next sprint, ensuring the platform evolved according to actual requirements.

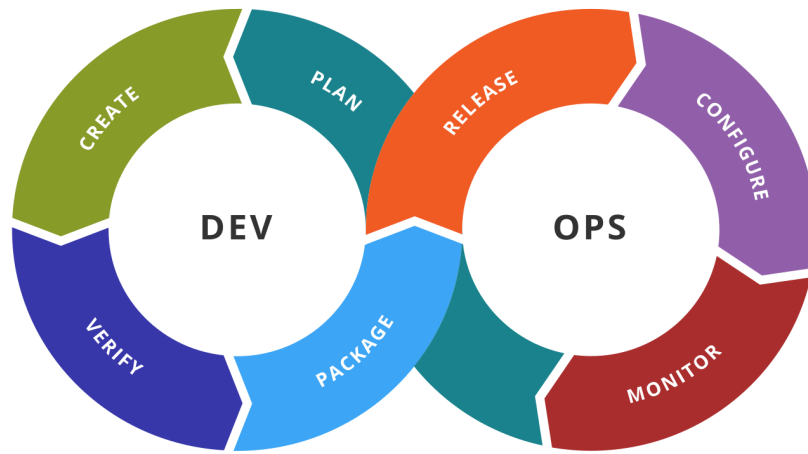


Figure 4.1: Devops

As illustrated in Figure 4.1, the DevOps lifecycle represents the continuous and iterative methodology adopted during the development of the CareerCompass platform. The diagram outlines the infinite loop of integration between development (DEV) and operations (OPS), encompassing key stages such as planning, creating, verifying, and packaging the software, seamlessly followed by releasing, configuring, and monitoring the system. Adopting this cyclical approach ensured continuous improvement, efficient code management, and overall system stability throughout the project's implementation phases.

4.7 System Interfaces and Results This section presents the graphical user interfaces (GUIs) of the CareerCompass platform and evaluates the system's operational results. The interfaces are designed using a modern tech stack to ensure high responsiveness, user-centric navigation, and seamless visual consistency across all functional modules.

4.7.1 The Home Page Interface The Home Page serves as the primary gateway to the CareerCompass platform. It establishes the visual identity of the system, utilizing an elegant layout featuring a clean background grid and dynamic, animated gradient blobs that elevate user engagement.

تبدیل المستخدم

لوحة التحكم

CareerCompass

THE FUTURE STARTS HERE

مرحباً بك،

Noor Ababneh

منصة التوجيه الأكاديمي الشاملة لطلاب الأردن

مسارك من المدرسة ...

إلى التخصص الجامعي الأنسب لك!

كيريير كومباس CareerCompass

رفيقك الأكاديمي الأول في مرحلة الانتقال إلى التعليم العالي: حيث تدمج بين التحليل العلمي للميول والتقييم الدقيق للأداء المدرسي. نساعدك على استكشاف التخصصات الجامعية وفهم متطلباتها، لنرسم لك مساراً تعليمياً متكاملًا يضمن لك مستقبلًا مبنياً على الشغف والتميز الدراسي.

خطة أكاديمية

مصفحة خصيصاً لك

4 حقول أكاديمية

حسب النظام الجديد

25+ جامعة أردنية

بيانات محدثة 2025

45+ مهارة مُقاسة

بمعايير عالمية

أدوات ذكية تقودك نحو القمة

بنينا المنصة لترافقك في كل خطوة: من فهم نفسك وحتى اختيار تخصصك الجامعي بكل ثقة.

الدليل الجامعي

قاعدة بيانات ضخمة تشمل الجامعات الأردنية، التخصصات، رسوم الساعات، والحدود الدنيا المحدثة للقبول الموحد.

تصفح الجامعات ←

التشخيص الأكاديمي

أدخل علامتك المدرسية الحالية لنكتشف الفجوات ونبني لك خطة دراسية دقيقة ترفع من فرص قبولك في الجامعات.

ابن خطتك ←

مقياس الميول المهنية

أجب عن مقياس (RIASEC) العلمي لتحلل نمط شخصيتك ونطابقه مع التخصصات التي تضمن لك شغفاً مستداماً.

ابدأ التقييم ←

4.7.2 Student Profiling and Onboarding Interface To deliver personalized academic guidance, the platform utilizes a streamlined data entry interface to establish the student's session profile.



→ العودة للرئيسية

CareerCompass
أدخل بياناتك لنبدأ رحلتك الأكاديمية

الاسم الكامل
Noor Ababneh

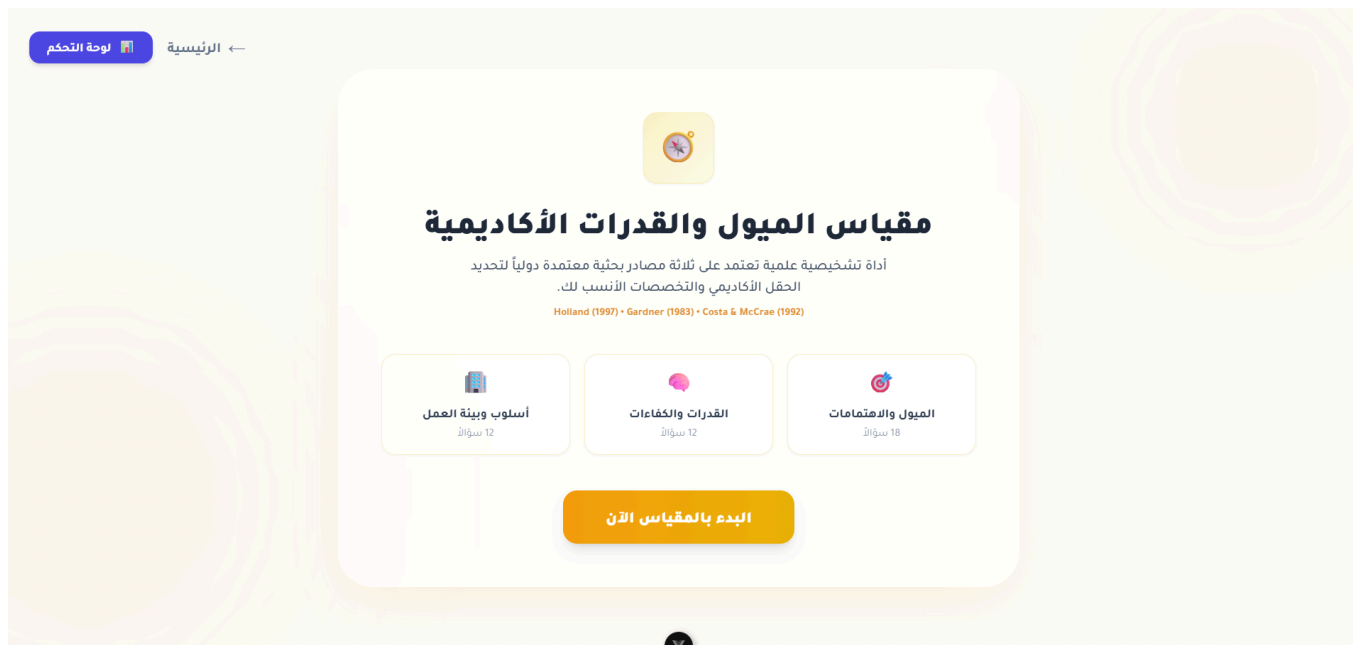
البريد الإلكتروني (اختياري)
noorababneh200@gmail.com

الجنس
☒ أنثى
 ☐ ذكر

المرحلة الدراسية
☐ الصف 10
 ☐ الصف 9
 ☒ الصف 11

التوجيهي

4.7.3 Multi-Dimensional Career Assessment Interface This module hosts the core psychometric evaluation tool, utilizing 42 carefully crafted scenarios distributed across three integrated analytical approaches (Interests, Work Styles, and Aptitudes).



لوحة التحكم الرئيسية ←

مقياس الميول والقدرات الأكاديمية
أداة تشخيصية علمية تعتمد على ثلاثة مصادر بحثية معتمدة دولياً لتحديد الحقل الأكاديمي والتخصصات الأنسب لك.
Holland (1997) • Gardner (1983) • Costa & McCrae (1992)

☒ أسلوب بيئة العمل 12 سؤالاً
☐ القدرات والكفاءات 12 سؤالاً
☐ الميول والاهتمامات 18 سؤالاً

البدء بالمقياس الآن

إعادة المقياس خطة التطوير الأكاديمي CareerCompass

تم اكتمال التحليل المنهجي بنجاح

بناءً على ميولك، الحقل الأنسب لك هو

حقل العلوم والتكنولوجيا والهندسة

هذا التحليل مبني على تقييم 42 معياراً دولياً لمطابقة الأنماط المهنية مع النظام التعليمي الأردني المحدث لعام 2024-2025.

سماتك الأكاديمية

بناءً على أنماطك: CE

منظم ومنهجي

قيادي طموح

منظم طموح: تجيد إنجاز الأهداف بكفاءة عالية، وتبرع في مجالات المحاسبة وإدارة المشاريع.

توزيع الأبعاد الشخصية

RIASEC PROFILE

CER

واقعي تقني 68%

تحليلي بحثي 66%

إبداعي فني 66%

4.7.4 Academic Diagnosis and Grades Input Interface The academic diagnosis component allows the platform to blend scientific interest-profiling with actual scholastic performance.

المنهج الأردني المحدث 2025/2026

سجل علاماتك الأكاديمية

أدخل درجاتك الحالية بدقة، ليقوم النظام بتحليل مستواك وتصميم خطة تطوير مخصصة تسد الفجوات وتضمن لك القبول الجامعي الذي تطمح إليه.

حدد مرحلتك الدراسية الحالية

تم التحديد من ملفك

مرحلة التأسيس

الصفوف 9 - 11

مرحلة التوجيه

الصف 12

ما هو حقلك الأكاديمي الحالي؟

الحقل الصحي

الطب - الأسنان - الصيدلة - التمريض

حقل العلوم والتكنولوجيا والهندسة

الهندسة - الحاسوب - الذكاء الاصطناعي

الوضع الأكاديمي: جيد جداً

وضع تنافسي قوي - في المنطقة الآمنة.

معدلك ممتاز ويؤكد لمجموعة واسعة من التخصصات المميزة. راجع التفاصيل في قسم التخصصات المرشحة أدناه.

80%

المعدل
التقريبي



تخصصات منافسة
+85%



قبول حكومي
+65%



قبول خاص
+65%

التخصصات المتاحة في حقلك

ترتبة حسب الحد الأدنى للقبول. يعتمد القبول الفعلي على التناقص

1 تمريض

حد القبول: 70% (حكومي: 70%)

✓ مؤهل

2 علوم طبية مخبرية

حد القبول: 70% (حكومي: 70%)

✓ مؤهل

3 العلوم الطبية المساندة: الأشعة، التخدير، العلاج الطبيعي، العلاج التنفسي

حد القبول: 70% (حكومي: 70%)

✓ مؤهل

4 دكتور صيدلة (PharmD)

حد القبول: 80% (حكومي: 80%)

⚠ تخصص تنافسي - المعدل الفعلي للقبول عادة أعلى من الحد الأدنى

✓ مؤهل

5 صيدلة

حد القبول: 80% (حكومي: 80%)

⚠ تخصص تنافسي - المعدل الفعلي للقبول عادة أعلى من الحد الأدنى

✓ مؤهل

6 طب بشري

حد القبول: 85% (حكومي: 85%)

غير مؤهل بعد

7 طب أسنان

حد القبول: 85% (حكومي: 85%)

غير مؤهل بعد

التشخيص التفصيلي للمواد

ترتبة من الأضعف للأقوى - تحليل الفجوة أكاديمياً

الكيمياء

جيد

72%

برنامج الإتيقان الأكاديمي

ACADEMIC MASTERY PROGRAM – المستوى: تنافسي / ما دون الامتياز

المستوى الحالي يؤهل لأغلب التخصصات – الهدف الوصول لمستوى الامتياز

المرحلة الأولى: تحليل الأداء المتقدم

الأسبوعان 2-1

تحديد نقاط الخسارة الخفية

✓ ابحث عن "الأسئلة الفخ" التي تبدو سهلة لكنها مصدر خسارة غير متوقعة.

✓ حلل امتحانات التدريب بمنهجية كمية؛ وثق كل خطأ ونوعه (مفاهيمي / إجرائي / إهمال).

✓ راجع أصعب أسئلة سنوات 2019-2024 واحص كم منها كنت ستجيب عنه بشكل كامل.

✓ حدد بدقة الفجوة بين أداء التدريب وأداء الامتحانات الرسمية - وحلل أسبابها.

✓ اسأل معلمك تحديداً: ما الأسئلة التي يراها الممتحنون متميزة؟ هذه المعلومة تساوي ذهباً.

✓ اعمل جدولاً: كل مادة x نوع الخطأ الأكثر تكراراً x الحل المقترح. هذا الجدول هو قاتلك.

أدوات الخطوة: [Wolfram Alpha](#) [سنوات وزارية 2020-2024](#)

1

المرحلة الثانية: الإتيقان تحت الضغط

الأسابيع 3-5

أداء >90% في كل تدريب

✓ حل كل خطأ صغير مهما بدا تافهاً؛ اكتب سببه وطريقة تجنبه مستقبلاً.

✓ امحن نفسك أسبوعياً بامتحان وزاري كامل بتوقيت أقصر من الرسمي (80% من الوقت).

2

4.7.5 Jordanian Universities Directory This interface serves as an integrated information hub containing comprehensive, up-to-date data regarding local higher education options.

خاصة
حكومية
الكل

جامعة حكومية

جامعة العلوم والتكنولوجيا (JUST)

إربد - الرمثا

أبرز الكليات

الطب
الهندسة
الصيدلة
المزيد >

تصفح التخصصات

جامعة حكومية

الجامعة الأردنية (JU)

عمان - الجيبة

أبرز الكليات

الطب
الهندسة
العلوم
المزيد >

تصفح التخصصات

جامعة حكومية

جامعة آل البيت

المفرق

أبرز الكليات

الشريعة
القانون
تكنولوجيا المعلومات
المزيد >

تصفح التخصصات

جامعة آل البيت

المفرق

الموقع الرسمي

التخصصات: 48

عدد التكنيات: 11

كلية الأمير الحسين بن عبدالله الثاني لتكنولوجيا المعلومات

عدد التخصصات المتاحة: 7

الكليات المتاحة

كلية الأمير الحسين بن عبدالله الثاني لتكنولوجيا المعلومات

كلية العلوم الطبية التطبيقية

كلية التمريض

كلية الهندسة

كلية الأعمال

كلية العلوم

علم الحاسوب (CS)

دراسة شاملة للبرمجة، الخوارزميات، وهياكل البيانات، مع التركيز على حل المشكلات برمجياً.

الساعات: 132

المجال: IT

مدة الدراسة: 4 سنوات

نظم المعلومات الصحية (HIS)

إدارة وتطوير الأنظمة التقنية المستخدمة في المستشفيات وقطاع الرعاية الصحية.

الساعات: 132

المجال: IT

مدة الدراسة: 4 سنوات

مجالات العمل

تطوير تطبيقات الأجهزة الذكية

تصميم وبرمجة تطبيقات الهواتف الذكية والأجهزة اللوحية (Android & IOS).

الساعات: 132

المجال: IT

مدة الدراسة: 4 سنوات

مجالات العمل المتاحة

مطور تطبيقات أندرويد

مطور تطبيقات IOS

مهندس برمجيات موبايل

مبرمج واجهات مستخدم

إلغاء مجالات العمل

الكليات المتاحة

كلية الأمير الحسين بن عبدالله الثاني لتكنولوجيا المعلومات

كلية العلوم الطبية التطبيقية

كلية التمريض

كلية الهندسة

كلية الأعمال

كلية العلوم

كلية الآداب والعلوم الإنسانية واللغات

كلية بيت الحكمة للعلوم السياسية

كلية العلوم التربوية

كلية الشريعة

4.7.6 Student Dashboard and Improvement Plan The Student Dashboard represents the ultimate outcome phase of the CareerCompass ecosystem, where analytical backend computations are translated into actionable insights.



4.8 Summary

In conclusion, this chapter provided a detailed overview of the implementation phase, technical stack, and testing procedures executed for the CareerCompass platform. The graphical user interfaces (GUIs) presented in the final section visually demonstrate the core features and services delivered by the system. These interfaces illustrate how the underlying diagnostic algorithms and weighted multi-dimensional logic are successfully translated into an intuitive, seamless, and user-friendly experience for high school students.

Chapter 5: Challenges and Difficulties Encountered

5.1 Introduction Developing an effective academic guidance system requires highly stable underlying data. However, during the software development lifecycle of CareerCompass, the team encountered complex systemic and technical obstacles. This chapter discusses the core difficulties faced, primarily driven by the dynamic and shifting nature of the Jordanian educational landscape.

5.2 The Shifting Landscape of the Jordanian Tawjihi System The most formidable challenge encountered during system design and database population was the inherent instability of the Jordanian secondary education system (Tawjihi). The Ministry of Education frequently implements structural updates, requiring continuous algorithmic and architectural adaptation:

- **Frequent Changes in Academic Fields and Prerequisites:** The constant updates from the Ministry of Education made it highly challenging to maintain a stable system. We initially designed and programmed the platform based on a six-field educational framework. However, sudden official decisions reduced this structure to only four fields. This major shift forced us to completely rebuild the core logic of our recommendation engine. In addition, the mandatory subjects required for each field changed multiple times during our work. These continuous changes made it difficult to finalize the project, as they forced us to repeatedly rewrite the calculation codes, adjust the gap analysis formulas, and update the database to ensure our system remained accurate and fully up-to-date with the latest rules.
- **Inter-Generational Shifts (Cohort Variations):** Requirements and passing criteria in Jordan often shift from one academic generation (Cohort) to the next. Different cohorts frequently face altered admission policies, dynamic prerequisite subjects, and fluctuating minimum acceptance GPAs. Designing a database schema

flexible enough to evaluate students accurately based on their specific cohort rules was highly complex.

- **Curriculum Updates and Subject Weighting:** The frequent introduction of new foundational subjects and ongoing changes to how individual subject grades are weighted to calculate the final Tawjihi average demanded a highly adaptable calculation engine capable of accommodating real-time formula adjustments.
- **Data Scarcity and Fragmentation:** Because ministerial decisions and specific university admission thresholds for newly introduced fields are updated dynamically, gathering consolidated, localized, and verified database entries proved exceptionally difficult and required continuous manual validation.

5.3 Algorithmic and Client-Side Integration Challenges

- **Multi-Model Weight Balancing:** Fine-tuning the mathematical integration between Holland's RIASEC professional interests, cognitive aptitudes, and work style stamina to output a singular, optimal field recommendation without introducing logical bias.
- **Asynchronous Flow Management:** Maintaining multi-step assessment questionnaires and persistent student grade context inside client-side Pinia stores smoothly while preventing UI freezing during asynchronous backend API fetches.

Chapter 6: Conclusion and Future Work

6.1 Conclusion The CareerCompass platform successfully addresses the acute pre-enrollment academic guidance gap within the Jordanian educational framework. By synthesizing psychological interest assessment with structured academic performance metrics, the project delivers a highly accessible digital advisory utility. Implementing custom multi-track logic successfully supports both early preparation via gap-analysis action plans for Grades 9–11 and immediate, data-driven "Top 3 Major" matching aligned directly with Jordan's new secondary fields for Grade 12 students. Ultimately, the system demonstrates how modern web architectures can empower youth to make confident career decisions while minimizing major-switching resource waste.

6.2 Future Work To expand the platform's technological capabilities and educational scope, several strategic enhancements are outlined for future iterations:

- **Cloud Deployment and Public Hosting:** Transitioning the locally built system to a robust cloud infrastructure (such as Netlify for the frontend and cloud-managed servers for the FastAPI backend) to ensure continuous public availability and scalable performance for students nationwide.
- **Generative AI Conversational Assistant:** Integrating an advanced Large Language Model (LLM) API to serve as an intelligent, round-the-clock interactive guidance chatbot answering custom student queries.
- **Unified Admissions System Integration:** Developing dynamic backend bridges to sync directly with Jordan's national unified admissions portal to present real-time seat availability and updated cut-off GPAs.
- **Native Mobile Application Releases:** Packaging the responsive web application into dedicated, standalone iOS and Android releases to provide enhanced offline access and continuous push notifications.

- **Comprehensive Longitudinal Tracking:** Implementing multi-year academic portfolio tracking to monitor skill development from Grade 9 through successful university graduation.

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